

Effective normal stress in HBI: governing equations, the `limitsigma` implementation, and a state-corrupting clamp

Notes for job 632510 (Cooper Basin / Habanero)

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Abstract

The cumulative-slip profile of job 632510 has a local *minimum* at the injection point rather than a maximum. This document (i) writes out the physics HBI solves and how it is discretised, (ii) shows that the `limitsigma` floor on effective normal stress is applied in a way that writes back into the stored state, so that the total normal stress ratchets upward without bound, (iii) verifies against the job 632510 output that the stored total normal stress equals $\sigma_{\min} + \max_t p$ to within 6×10^{-3} MPa in every cell that ever hit the floor, and (iv) explains why a 115 MPa error in normal stress produces only a 1.1 cm signature in slip. A patch is proposed in §8.

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1 Scope and conventions

Everything below refers to **problem 3dp**: a planar rectangular fault discretised into $i_{\max} \times j_{\max}$ uniform square cells, with along-fault fluid-pressure diffusion switched on (`pressurediffusion T`). Line numbers refer to the working tree at the time of writing.

HBI uses a mixed unit system; getting this right matters for reading the source. Table 1 lists the conventions actually used in `main_LH.f90` and `m_diffusion.f90`.

Quantity	Unit	Notes
stress (τ, σ, p)	MPa	<code>sigmainit 30.0</code> is 30 MPa
length (<code>ds, xcol</code>)	km	<code>ds 0.005</code> is a 5 m cell
slip	m	
slip rate	m/s	<code>vref</code> = 10^{-6} (1.150)
time	s	<code>tmax</code> is in years, converted on read
rigidity <code>rigid</code>	GPa	<code>rigid 24</code>
shear wave speed <code>vs</code>	km/s	default 3.464
permeability <code>kp</code>	m^2	
viscosity <code>eta</code>	Pa s	
compressibility <code>beta</code>	Pa^{-1}	

Table 1: Unit conventions. The mixed GPa/km/MPa choice is self-consistent: e.g. `rigid/vs` has units $\text{GPa}/(\text{km s}^{-1}) = \text{MPa}/(\text{m s}^{-1})$, which is the correct unit for the radiation-damping coefficient.

Throughout, $\bar{\sigma}$ denotes *effective* normal stress and Σ *total* normal stress, related by Terzaghi’s principle,

$$\bar{\sigma} = \Sigma - p. \quad (1)$$

In the code these are the arrays `sigma` and `sigmat` respectively, and `pf` is p . Note that `sigma` — despite the name — is the *effective* normal stress; this is the single most common source of confusion when reading `main_LH.f90`.

2 Governing equations

2.1 Quasi-dynamic elasticity

The fault is embedded in a homogeneous elastic half-space. Slip $\delta(\mathbf{x}, t)$ on the fault generates shear and normal traction changes given by a boundary integral over the fault surface. Discretised on N

cells, with $V_j = \dot{\delta}_j$,

$$\dot{\tau}_i^{\text{el}} = \sum_{j=1}^N K_{ij}^s (V_j - V_{\text{pl}}), \quad \dot{\Sigma}_i = \sum_{j=1}^N K_{ij}^n (V_j - V_{\text{pl}}), \quad (2)$$

where K^s and K^n are the shear-shear and shear-normal stress-interaction kernels. For rectangular cells these come from the Okada (1992) dislocation solution (`okada.f90`, reached through `HACApK.entry_ij` in `m_HACApK.calc_entry_ij.f90:61--63`). The V_{pl} subtraction is the backslip formulation; this run sets `vp1 0d0`, so there is no tectonic loading and all stressing comes from the injection.

The dense matrices K^s, K^n are never formed. They are compressed into lattice $\mathcal{H}$ -matrices by `HACApK` and applied as matrix-vector products (`HACApK.adot_lattice_hyp`, `main_LH.f90:1852,1863`).

A planar fault has no self-induced normal stress change. For a planar fault, slip on the plane produces *zero* change in the normal traction on that same plane: $K^n \equiv 0$. The code recognises this explicitly and skips the normal kernel entirely (`main_LH.f90:501--503`):

```
select case(problem)
case('2dp','2dn3','3dp','2dvs')
  sigmaconst=.true.
end select
```

so that for `3dp` the code sets $\dot{\Sigma}^{\text{el}} = 0$ (`main_LH.f90:1860--1861`). **This is important for what follows: in this run the total normal stress has no elastic contribution at all, so Σ must remain exactly $\sigma_{\text{init}} = 30$ MPa for all time, and the effective normal stress must be exactly $30 - p$.**

2.2 The quasi-dynamic approximation

Full elastodynamics is replaced by the radiation-damping approximation: the inertial term is modelled by a local dashpot,

$$\tau = \tau^{\text{el}} - \eta V, \quad \eta = \frac{\mu}{2c_s}, \quad (3)$$

with μ the shear modulus and c_s the shear wave speed. In the code η appears as `0.5d0*rigid/vs`.

2.3 Rate-and-state friction

Sliding obeys a regularised rate-and-state law. HBI carries the state as ψ , a friction-like variable, and inverts the friction law for slip rate (`main_LH.f90:1820`):

$$V = 2V_{\text{ref}} \exp\left(-\frac{\psi}{a}\right) \sinh\left(\frac{\tau}{a\bar{\sigma}}\right). \quad (4)$$

This is the standard regularisation of $\tau = \bar{\sigma} [f_0 + a \ln(V/V_0) + b \ln(V_0\theta/d_c)]$, well behaved as $V \rightarrow 0$ and for $V < 0$.

The state evolves by the ageing law (`evlaw` defaults to `aging`; `main_LH.f90:1924--1929`):

$$\dot{\psi} = \frac{bV_{\text{ref}}}{d_c} \exp\left(\frac{f_0 - \psi}{b}\right) - \frac{b|V|}{d_c}. \quad (5)$$

This is exactly $\dot{\theta} = 1 - V\theta/d_c$ rewritten under the change of variable $\psi = f_0 + b \ln(V_{\text{ref}}\theta/d_c)$: differentiating gives $\dot{\psi} = (b/\theta)\dot{\theta} = b/\theta - bV/d_c$, and $b/\theta = (bV_{\text{ref}}/d_c) \exp((f_0 - \psi)/b)$.

Job 632510 uses $a = 0.015$, $b = 0.012$, so $a - b > 0$: the fault is *velocity-strengthening* everywhere and slip is aseismic throughout.

2.4 The evolution equation for shear stress

Differentiating (3) and using $V = V(\psi, \tau, \bar{\sigma})$ from (4),

$$\dot{\tau} = S - \eta \left(\frac{\partial V}{\partial \psi} \dot{\psi} + \frac{\partial V}{\partial \tau} \dot{\tau} + \frac{\partial V}{\partial \bar{\sigma}} \dot{\bar{\sigma}} \right), \quad S \equiv \dot{\tau}^{\text{el}}, \quad (6)$$

which solved for $\dot{\tau}$ gives, with $\dot{\bar{\sigma}} = \dot{\Sigma} - \dot{p}$,

$$\dot{\tau} = \frac{S - \eta \left[\frac{\partial V}{\partial \psi} \dot{\psi} + \frac{\partial V}{\partial \bar{\sigma}} (\dot{\Sigma} - \dot{p}) \right]}{1 + \eta \frac{\partial V}{\partial \tau}} \quad (7)$$

with partial derivatives obtained from (4),

$$\frac{\partial V}{\partial \tau} = \frac{2V_{\text{ref}}e^{-\psi/a}}{a\bar{\sigma}} \cosh \frac{\tau}{a\bar{\sigma}}, \quad \frac{\partial V}{\partial \bar{\sigma}} = -\frac{2V_{\text{ref}}e^{-\psi/a}\tau}{a\bar{\sigma}^2} \cosh \frac{\tau}{a\bar{\sigma}}, \quad \frac{\partial V}{\partial \psi} = -\frac{V}{a}. \quad (8)$$

Equation (7) is `main_LH.f90:1963` verbatim; the partial derivatives are lines 1960–1962. Note that the pore-pressure rate $\dot{p}$ enters only here, as a correction to the effective-stress rate.

2.5 Fluid pressure diffusion

Along-fault pressure obeys a linear diffusion equation with a heterogeneous, slip-dependent diffusivity and a well source (`m_diffusion.f90`, `diffusion3dwop/Beuler2d`):

$$\beta\phi \frac{\partial p}{\partial t} = \nabla \cdot \left(\frac{k}{\eta_f} \nabla p \right) + q, \quad c_{\text{hyd}} = \frac{k}{\eta_f \beta \phi}, \quad (9)$$

implemented as `cdiff = kpG/(eta*beta*phiG)*1d-6` (`m_diffusion.f90:443--445`; the 10^{-6} converts m^2/s to km^2/s to match `ds0` in km). The scheme is backward Euler in time, second order in space, solved with conjugate gradients; boundary conditions are set by `bcl/bcr/bct/bcb` (all Dirichlet, $p = 0$, in this run).

Permeability evolution. With `permev T` (`main_LH.f90:2402--2405`),

$$\frac{dk}{dt} = -\frac{V}{k_L} (k - k_{\text{max}}) - \frac{k - k_{\text{min}}}{k_T}, \quad (10)$$

i.e. slip enhances permeability toward k_{max} over a slip scale k_L , and it heals back toward k_{min} over a time scale k_T . Job 632510 uses $k_{\text{min}} = 10^{-15}$, $k_{\text{max}} = 5 \times 10^{-14} \text{ m}^2$, $k_L = 10^{-3} \text{ m}$, $k_T = 10^{15} \text{ s}$ (i.e. healing effectively disabled). The corresponding hydraulic diffusivities are $c_{\text{hyd}} \approx 5.6 \times 10^{-3} \text{ m}^2/\text{s}$ (intact) to $0.28 \text{ m}^2/\text{s}$ (fully enhanced).

Well source. The injector is a single cell with a Peaceman well-index coupling to a wellbore of radius r_w with storage S_w (`m_diffusion.f90:669--675`):

$$T = \frac{2\pi k}{\eta_f [\ln(0.2 \Delta s / r_w) + s_{\text{skin}}]}, \quad \gamma = \frac{S_w / \Delta t}{S_w / \Delta t + T}, \quad (11)$$

with the wellbore pressure updated as $p_w^{n+1} = \gamma(p_w^n + \Delta t q / S_w) + (1 - \gamma)p_{\text{cell}}^{n+1}$ (`m_diffusion.f90:763`). The injection schedule $q(t)$ is read from `june_clean.txt` and linearly interpolated between breakpoints.

2.6 The coupled system

Per cell the state vector is $y = (\psi, \tau, \bar{\sigma})$, and the code integrates

$$\frac{d}{dt} \begin{pmatrix} \psi \\ \tau \\ \bar{\sigma} \end{pmatrix} = \begin{pmatrix} (5) \\ (7) \\ \dot{\Sigma}^{\text{el}} \end{pmatrix}, \quad (12)$$

with an adaptive fifth-order Cash–Karp Runge–Kutta step (`rkqs`, `main_LH.f90:1979`). Note the third component: within an RK step $\bar{\sigma}$ is advanced by the *elastic* normal-stress rate only. Pore pressure is not integrated by the RK step at all — it is handled by operator splitting, described next.

3 How a timestep is assembled

One iteration of the main loop (`main_LH.f90:1090--1135`) does:

1. **RK step.** `rkqs` advances y over Δt using (12). Pore pressure is frozen at its old value; only $\dot{p}$ from the previous step enters, through (7). (`main_LH.f90:1091`)
2. **Unpack and accumulate slip.** $\psi, \tau, \bar{\sigma}$ are copied out of y ; slip is accumulated with a trapezoidal rule. (`main_LH.f90:1102--1122`)
3. **Reconstruct total normal stress.** (`main_LH.f90:1124`)

```
sigmat(i)=sigma(i)+pf(i)
```

4. **Diffusion step.** `pressure_diffusion` advances (10) and (9) over the same Δt , then re-imposes Terzaghi’s principle and applies the clamp (`main_LH.f90:2449--2458`):

```
pf(i)=pfG(i_)
dpdt(i)=(pf(i)-pfo(i))/dtdid
sigma(i)=sigmat(i)-pf(i)
if(limitsigma) then
  if(sigma(i)<minsig) sigma(i)=minsig
  if(sigma(i)>maxsig) sigma(i)=maxsig
end if
y(3*i)=sigma(i)
```

The intent of the splitting is clear and correct: elasticity moves Σ , diffusion moves p , and $\bar{\sigma} = \Sigma - p$ is reassembled once per step. The defect is in how the clamp interacts with step 3.

4 The limitsigma clamp

4.1 What it is for

Effective normal stress appears in the denominator of (4). If $\bar{\sigma} \rightarrow 0$ the argument of $\sinh$ diverges; if $\bar{\sigma} < 0$ the friction law is meaningless (the fault would be in tension and should open). Runs in which injection drives p toward or past the ambient normal stress therefore need some regularisation. HBI provides two:

- **opening** T: a physical treatment in which $\bar{\sigma} < \sigma_{\min}$ opens the fault, generating opening displacement that feeds back through the normal kernel (`main_LH.f90:1838--1841`). This forces `sigmaconst=.false.` even for `3dp`.
- **limitsigma** T (the default, and what job 632510 uses): a numerical floor, $\bar{\sigma} \leftarrow \max(\bar{\sigma}, \sigma_{\min})$.

A floor of this kind is defensible as a numerical safeguard *provided it is a pure projection*: it should modify the value handed to the friction law without altering the underlying stress state. Under that reading, the correct result is

$$\bar{\sigma}(t) = \max\left(\sigma_{\text{init}} + \Delta\Sigma^{\text{el}}(t) - p(t), \sigma_{\min}\right), \quad (13)$$

which is a *memoryless* function of the current p : as soon as p falls again, $\bar{\sigma}$ comes back up along $\sigma_{\text{init}} - p$.

There is also a second, separate guard inside `deriv` (`main_LH.f90:1912--1916`) that zeroes $\dot{\Sigma}^{\text{el}}$ when $\bar{\sigma}$ is outside $[\sigma_{\min}, \sigma_{\max}]$. That one is benign: it suppresses a rate, it does not rewrite a state.

4.2 The defect

The clamped value written into `y(3*i)` at step 4 is read back at step 2 of the *next* iteration and then, at step 3, added to p to reconstruct Σ . The clamp correction is therefore absorbed into the total normal stress and becomes permanent. HBI has no independent record of Σ against which to check this: `sigmat` is a derived quantity recomputed from `sigma` every step, not a state variable.

5 Derivation of the ratchet

Index the operator-split steps by n , and for a single cell write $\bar{\sigma}_n, p_n, \Sigma_n$ for the values of `sigma`, `pf`, `sigmat`. Let $\Delta_n = \int_{t_{n-1}}^{t_n} \dot{\Sigma}^{\text{el}} dt$ be the elastic increment accumulated by RK step n (identically zero for `3dp`, since `sigmaconst=.true.`).

The four operations of §3 are

$$(a) \text{ RK: } \tilde{\bar{\sigma}}_n = \bar{\sigma}_{n-1} + \Delta_n \quad (l. 1105) \quad (14)$$

$$(b) \text{ reconstruct: } \Sigma_n = \tilde{\bar{\sigma}}_n + p_{n-1} \quad (l. 1124) \quad (15)$$

$$(c) \text{ diffuse: } p_{n-1} \rightarrow p_n \quad (16)$$

$$(d) \text{ re-impose: } \bar{\sigma}_n^{\text{raw}} = \Sigma_n - p_n \quad (l. 2452) \quad (17)$$

$$(e) \text{ clamp: } \bar{\sigma}_n = \text{clamp}(\bar{\sigma}_n^{\text{raw}}) \quad (l. 2454-2455) \quad (18)$$

Composing (a), (b), (d):

$$\bar{\sigma}_n^{\text{raw}} = \bar{\sigma}_{n-1} + \Delta_n - (p_n - p_{n-1}). \quad (19)$$

Define the clamp excess $e_n \equiv \bar{\sigma}_n - \bar{\sigma}_n^{\text{raw}}$, which is zero on unclamped steps and positive whenever the floor bites. Then $\bar{\sigma}_n = \bar{\sigma}_{n-1} + \Delta_n - (p_n - p_0) + e_n$, and telescoping from $n = 0$:

$$\boxed{\bar{\sigma}_N = \underbrace{\sigma_{\text{init}} + \sum_{n \leq N} \Delta_n - (p_N - p_0)}_{\text{correct}} + \underbrace{\sum_{n \leq N} e_n}_{\text{accumulated clamp defect}}} \quad (20)$$

Compare with the intended behaviour (13). The correct result depends only on the *current* p ; the implemented result carries the sum of every clamp correction ever applied. That sum never decays.

5.1 Closed form for a planar fault

Set $\Delta_n \equiv 0$ (true for 3dp). Then from (b), $\Sigma_n = \bar{\sigma}_{n-1} + p_{n-1}$, and two cases arise:

- *Step $n - 1$ was not clamped.* Then $\bar{\sigma}_{n-1} = \Sigma_{n-1} - p_{n-1}$, so $\Sigma_n = \Sigma_{n-1}$: unchanged, as it should be.
- *Step $n - 1$ hit the floor.* Then $\bar{\sigma}_{n-1} = \sigma_{\min}$ and $\Sigma_n = \sigma_{\min} + p_{n-1}$. But hitting the floor means $\Sigma_{n-1} - p_{n-1} < \sigma_{\min}$, i.e. $\sigma_{\min} + p_{n-1} > \Sigma_{n-1}$. So Σ has *strictly increased*.

Σ is therefore non-decreasing, and increases exactly at the pressure peaks that trigger the floor. Hence, for any cell in which the floor was ever reached,

$$\boxed{\Sigma_{\text{final}} = \sigma_{\min} + \max_t p(t)} \quad (21)$$

and for a cell that never reached the floor, $\Sigma_{\text{final}} = \sigma_{\text{init}}$ exactly. Equivalently, the effective normal stress reported by the code after the pressure has receded is

$$\bar{\sigma}_{\text{code}}(t) = \sigma_{\min} + \max_{t' \leq t} p(t') - p(t) \quad \text{instead of} \quad \bar{\sigma}_{\text{true}}(t) = \max(\sigma_{\text{init}} - p(t), \sigma_{\min}). \quad (22)$$

The floor bites when $p > \sigma_{\text{init}} - \sigma_{\min}$, which for job 632510 is $p > 29$ MPa.

6 Verification against job 632510

Run `docs/verify_clamp_ratchet.py` to regenerate everything in this section. Grid 601×601 , $\Delta s = 5$ m, $\sigma_{\text{init}} = 30$ MPa, $\sigma_{\min} = 1$ MPa, injector at Fortran (301, 301), 1402 output frames spanning 30.66 days.

6.1 The ratchet identity

Table 2 tests (21) cell by cell along a profile through the injector.

The agreement is not approximate — it is the identity derived in §5, holding to five significant figures, with a sharp transition at exactly the predicted threshold. The clamped region is $|r| \leq 190$ m.

At the injector the consequence is severe: p peaks at 144 MPa against an ambient normal stress of 30 MPa, so Σ ends at 145 MPa and the reported effective normal stress reaches 135 MPa — a 115 MPa error, more than four times the true total normal stress.

j	r (m)	$\max_t p$	Σ_{final}	$\sigma_{\text{min}} + \max_t p$
300	0	144.00	145.01	145.00
305	25	71.12	72.12	72.12
310	50	55.61	56.61	56.61
317	85	44.04	45.04	45.04
325	125	36.51	37.51	37.51
340	200	28.47	30.00	29.47
360	300	21.79	30.00	22.79
400	500	14.22	30.00	15.22
450	750	10.02	30.00	11.02

Table 2: Stress in MPa. Cells with $\max_t p > 29$ MPa satisfy (21); cells that never reached the floor retain $\Sigma = \sigma_{\text{init}} = 30.0000$ exactly (bold). Maximum residual $|\Sigma - (\sigma_{\text{min}} + \max_t p)|$ over all clamped cells: 5.8×10^{-3} MPa, attributable to output frames subsampling the timesteps, so $\max_t p$ over frames slightly underestimates the true maximum.

6.2 Effect on slip

Because the ratchet only manifests when p *falls* — when p is at its running maximum the code and the truth agree, both sitting at σ_{min} — the injector cell is repeatedly and spuriously strengthened during the troughs of the injection schedule. At day 0.956, for instance, $p = 91$ MPa at the injector, so the true effective normal stress is pinned at 1 MPa; the code instead reports 29.08 MPa. The friction coefficient $\tau/\bar{\sigma}$ required to sustain slip collapses, and the cell locks: V drops from $\sim 10^{-6}$ to $\sim 10^{-24}$ m/s while its neighbours 85 m away continue sliding at $\sim 10^{-7}$ m/s.

Over the run, the injector cell spends 67.5% of output frames with an effective normal stress too high by more than 0.5 MPa; that fraction falls to 36.8% at $r = 50$ m, 16.3% at $r = 85$ m, and 0% beyond 200 m.

7 Why a 115 MPa error yields only a 1 cm cusp

This is the natural objection to the diagnosis, and it is worth answering carefully: the error in $\bar{\sigma}$ is two orders of magnitude larger than the stress scale of the problem, yet the visible artefact in Figure 24 is about one centimetre out of twenty.

7.1 The error is a switch, not a forcing

Once $\tau/\bar{\sigma}$ falls below the frictional strength, the cell is locked and V is exponentially small. It makes no difference whether $\bar{\sigma}$ is too high by 20 MPa or by 115 MPa: the cell is equally dead either way. The error therefore acts as an on/off switch on locking, and what it controls is the *duration* of the lock-ups, not their depth.

7.2 Elasticity bounds the lag

A patch of radius a that lags its surroundings by a slip deficit δ develops a stress concentration of order

$$\Delta\tau \sim \frac{\mu\delta}{a}. \quad (23)$$

With $\mu = 24$ GPa, $a \approx 100$ m and $\delta = 1.1$ cm this is $\Delta\tau \approx 2.6$ MPa. The entire stress-drop budget of this problem is about 10 MPa (initial shear stress $\mu_{\text{init}}\sigma_{\text{init}} = 0.37 \times 30 = 11.1$ MPa, falling to ~ 1 MPa in the slipped region). A centimetre-scale deficit already consumes a quarter of that. The surroundings then load the locked patch hard enough that it re-slips the moment p rises again and the clamp returns $\bar{\sigma}$ to σ_{min} . The fault simply cannot express a much larger lag.

7.3 Direct accounting

Splitting the growth of the injector-vs-85 m slip deficit by whether the injector's normal stress was corrupted at that instant:

Deficit accumulated while $\bar{\sigma}$ corrupted (> 0.5 MPa error)	+3.534 cm
Deficit recovered while $\bar{\sigma}$ correct	−2.429 cm
Net deficit (the depth of the cusp)	+1.105 cm

So the cusp is the residual of a repeated lock/catch-up cycle: the bug injects 3.5 cm of deficit and elastic loading heals 2.4 cm of it. This also explains the shape — the profile is smooth rather than showing a one-cell notch, because the healing phases redistribute the lag over the whole 190 m clamped patch.

7.4 The plot understates the error

Before the ratchet takes hold (up to day ~ 0.9), the injector correctly *leads* the 85 m cell by 2.8 cm, which is the physically expected behaviour. The ratchet erases that lead and then reverses it. The total error in relative slip is therefore about 4 cm on a 20 cm signal — roughly 20% — of which the figure shows only the 1.1 cm that ended up on the wrong side of zero.

8 Proposed fix

Σ must become a genuine state variable, advanced only by the elastic increment and never reconstructed from a clamped $\bar{\sigma}$. The code already uses this pattern in the analytical-injection path (`main_LH.f90:2470`, `sigma0(i)=sigma0(i)+y(3*i)-sigma(i)`); the numerical-diffusion path is the one that lacks it.

Introduce one extra array, `sigmac`, holding the value of $\bar{\sigma}$ that was written into `y` at the end of the previous diffusion step. The difference `sigma(i)-sigmac(i)` is then exactly Δ_n , the elastic increment of the RK step, uncontaminated by the clamp.

1. Declare and allocate `sigmac` alongside `sigmat` (`main_LH.f90:68,549`).
2. Initialise both once, in the loop that sets up `y` just before the time loop (`main_LH.f90:1078--1084`); this covers the fresh-start and restart paths alike, since both have `sigma` and `pf` populated by that point:

```
do i=1,NCELL
  y(3*i-2) = psi(i)
  y(3*i-1) = tau(i)
  y(3*i)   = sigma(i)
  sigmat(i) = sigma(i) + pf(i)    ! NEW: total normal stress, a true
                                state
```

```

    sigmac(i) = sigma(i)           ! NEW: last clamped value handed to
    the RK
end do

```

3. **Delete** the per-step reconstruction at `main_LH.f90:1124`:

```

    mu(i)=tau(i)/sigma(i)
-    sigmat(i)=sigma(i)+pf(i)

```

4. Advance `sigmat` by the elastic increment only, in `pressure_diffusion` (`main_LH.f90:2449--2458`):

```

do i=1,ncell
    i_=st_sum%lodec(i)
    pf(i)=pfG(i_)
    dpdt(i)=(pf(i)-pfo(i))/dtdid
    sigmat(i)=sigmat(i)+(sigma(i)-sigmac(i)) ! NEW: elastic
    increment only
    sigma(i)=sigmat(i)-pf(i)
    if(limitsigma) then
        if(sigma(i)<minsig) sigma(i)=minsig
        if(sigma(i)>maxsig) sigma(i)=maxsig
    end if
    sigmac(i)=sigma(i) ! NEW: remember what
    we handed out
    y(3*i)=sigma(i)
end do

```

With this change the clamp becomes the pure projection (13): e_n no longer enters (20), and for 3dp the identity $\bar{\sigma} = \max(\sigma_{\text{init}} - p, \sigma_{\text{min}})$ holds exactly for all time. The guard in `deriv` remains consistent: when $\bar{\sigma}$ sits at the floor, $\dot{\Sigma}^{\text{el}}$ is zeroed, so $\Delta_n = 0$ and `sigmat` is correctly left unchanged.

Regression check. For any 3dp run, assert $\max_i |\bar{\sigma}_i + p_i - \sigma_{\text{init},i}| < \epsilon$ at every output step. It should be zero to round-off, since `sigmaconst` is `.true.` and nothing may move the total normal stress.

Restart caveat. On restart, `sigma` is read back from `output/sigma*.dat` (`main_LH.f90:674--681`), so a restart from a run made with the current code inherits whatever corruption is already in that field. Restarted chains should be rerun from $t = 0$ after the fix.

9 Implications

Which results are affected. Any run in which p exceeded $\sigma_{\text{init}} - \sigma_{\text{min}}$ anywhere on the fault. For job 632510 that is a 190 m radius disc around the injector — which is the near-well region that most of the validation figures are about. Runs where p stayed below that threshold are untouched, exactly and verifiably (Table 2, bold entries).

A separate modelling question. Even with the ratchet fixed, this run drives p to 144 MPa against $\sigma_{\text{init}} = 30$ MPa, so everything above 29 MPa is discarded by the floor and the near-well effective stress is set by σ_{min} rather than by the physics. That is worth discussing on its own terms:

- The peak is a single 5 m cell acting as a point source, so its magnitude is grid-dependent; the Peaceman well index mitigates but does not remove this.
- $\sigma_{\text{init}} = 30$ MPa with p reaching several times that is outside the regime where a floored- $\bar{\sigma}$ model is meaningful. If the fault genuinely reaches $\bar{\sigma} \rightarrow 0$, the physical treatment is **opening T** (which also turns the normal kernel back on), not a numerical floor.
- Alternatively the injection schedule, wellbore storage S_w , or k_{min} may need revisiting so that p stays below σ_{init} .

The figure itself is fine. The cusp is present in the raw `slip632510.dat`; cell 85 of `cooper_basin_validation_stage_june15.ipynb` cuts through $j = \lfloor j_{\text{max}}/2 \rfloor = 300$, which is the injector column, and reproduces the binary data faithfully. One cosmetic note: cell 85 labels the i axis “down-dip” while cell 86 fixes i and sweeps j under the same label. The geometry is symmetric here so nothing follows from it, but the two cells disagree about which index is down-dip.

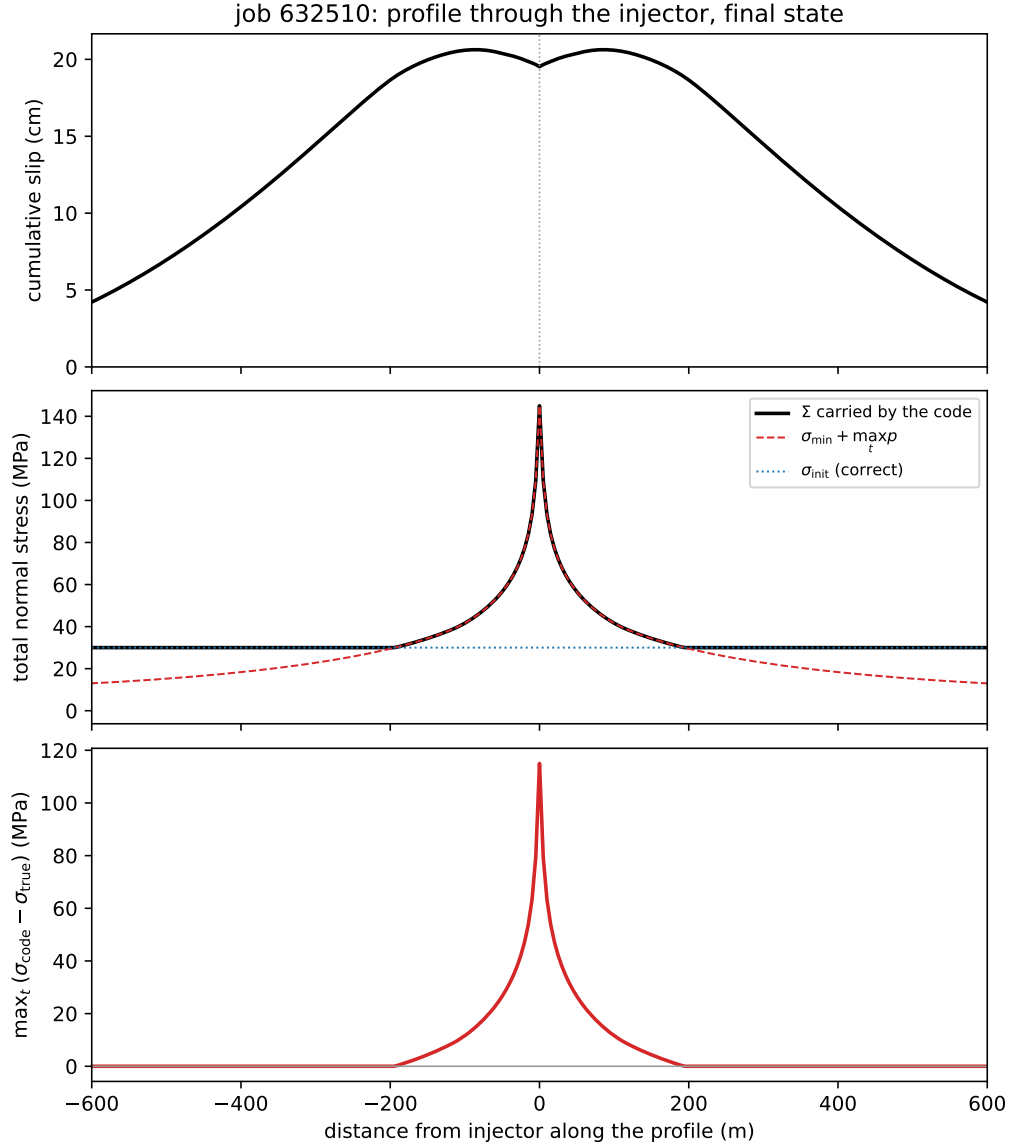


Figure 1: Profile through the injector at the final timestep. Top: the cumulative-slip cusp. Middle: total normal stress carried by the code (black) against the prediction (21) (red dashed) and the correct value (blue dotted). Bottom: peak error in effective normal stress. The error vanishes at $|r| \approx 190$ m, the edge of the region where p ever exceeded 29 MPa.

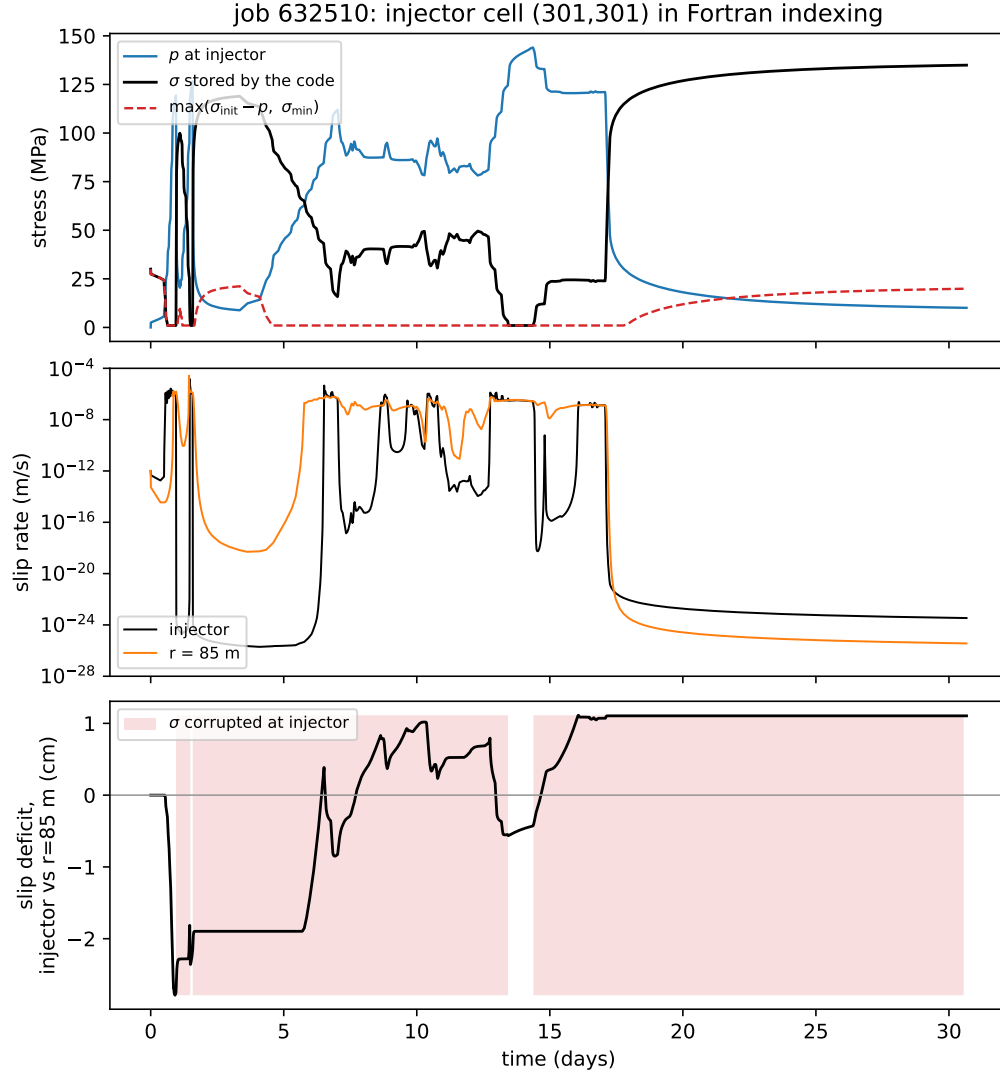


Figure 2: Time histories at the injector cell. Top: pore pressure, the stored effective normal stress, and the value required by (13). The two diverge from day ~ 1 onward, every time p recedes from a peak. Middle: slip rate at the injector versus a cell 85 m away — the injector switches off entirely during the corrupted intervals. Bottom: cumulative slip deficit between the two cells, with the corrupted intervals shaded.